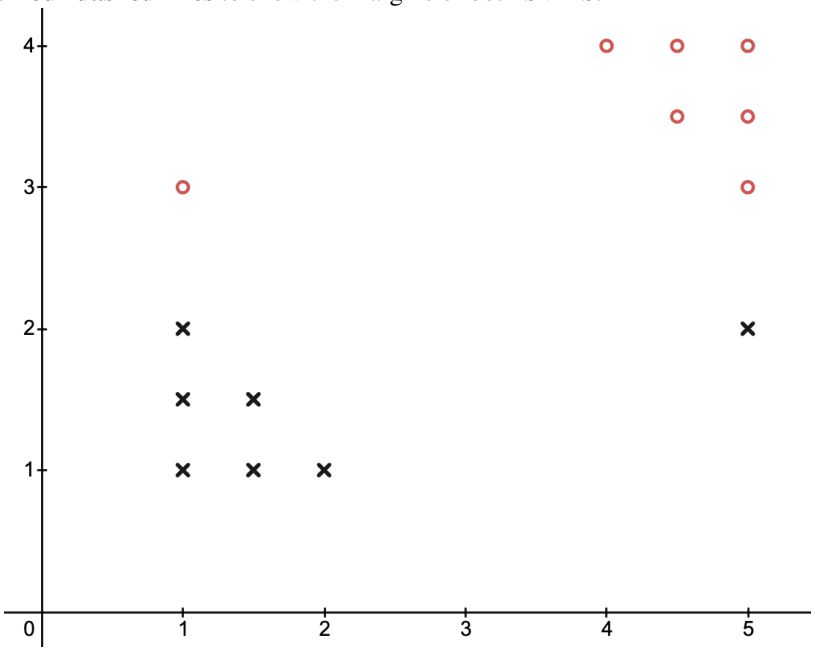
Q3. [10 pts] Decision Boundaries

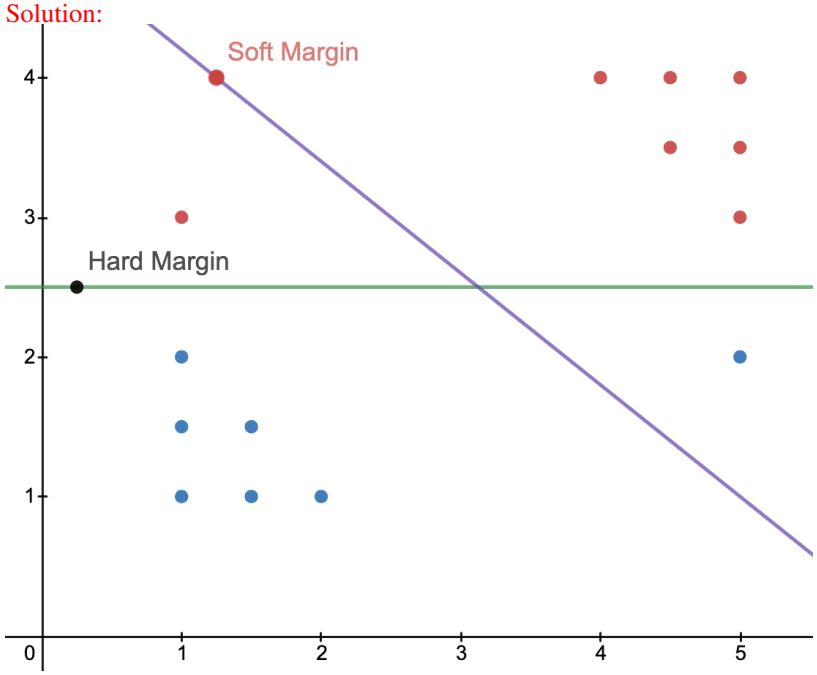
In the question, you will draw the decision boundaries that classifiers would learn.

(a) [6 pts] Given the sample points below, draw and label **two lines**: the decision boundary learned by a hard-margin SVM and the decision boundary learned by a soft-margin SVM. We are not specifying the hyperparameter C, but don't make C too extreme. (We are looking for a qualitative difference between hard- and soft-margin SVMs.) **Label the two lines clearly.**

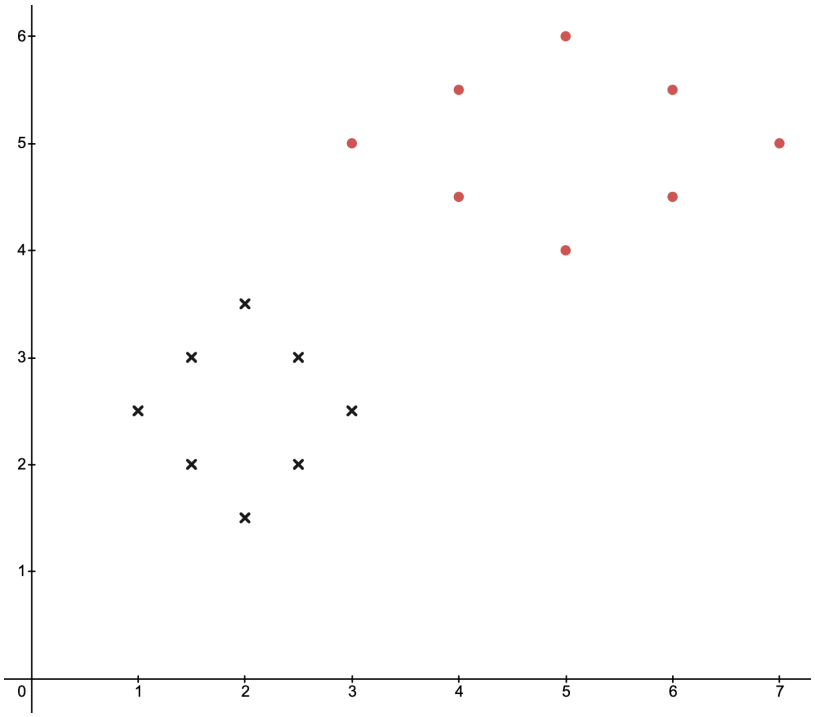
Also draw and label **four dashed lines** to show the margins of both SVMS.



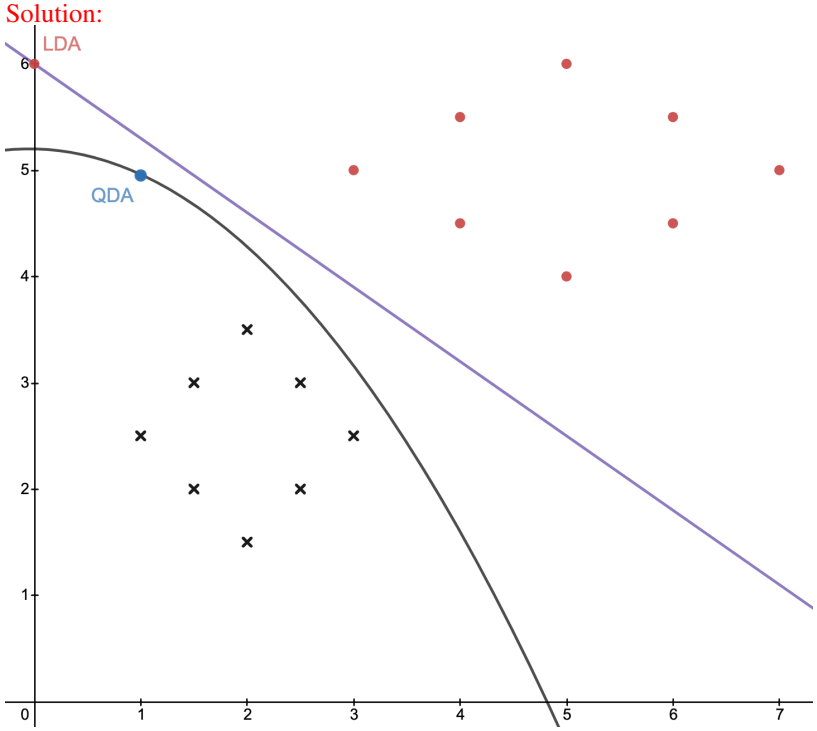
Solution:



(b) [4 pts] Given the sample points below, draw and label **two curves**: the decision boundary learned by LDA and the decision boundary learned by QDA. Label the two curves clearly.



Solution:



Q4. [16 pts] Kernel Principal Components Analysis

Let X be an $n \times d$ design matrix. Suppose that X has been centered, so the sample points in X have mean zero. In this problem we consider kernel PCA and show that it equates to solving a *generalized Rayleigh quotient problem*.

(a) [1 pt] Fill in the blank: every principal component direction for X is an eigenvector of _____. $X^\top X$

(b) [1 pt] Fill in the blank: an optimization problem can be kernelized only if its solution w is always a linear combination of the sample points. In other words, we can write it in the form $w =$ _____.
$X^\top a$ (for some vector a).

(c) [4 pts] Show that every principal component direction w with a nonzero eigenvalue is a linear combination of the sample points (even when $n < d$).

As w is an eigenvector of $X^\top X$, there is a $\lambda \in \mathbb{R}$ such that $X^\top X w = \lambda w$. Setting $a = \frac{1}{\lambda} X w$, we have $X^\top a = w$.

(d) [4 pts] Let $\Phi(z)$ be a feature map that takes a point $z \in \mathbb{R}^d$ and maps it to a point $\Phi(z) \in \mathbb{R}^D$, where D might be extremely large or even infinite. But suppose that we can compute the kernel function $k(x, z) = \Phi(x) \cdot \Phi(z)$ much more quickly than we can compute $\Phi(x)$ directly. Let $\Phi(X)$ be the $n \times D$ matrix in which each sample point is replaced by a featurized point. By our usual convention, row i of X is $X_i^\top$, and row i of $\Phi(X)$ is $\Phi(X_i)^\top$.

Remind us: what is the kernel matrix K? Answer this two ways: explain the relationship between K and the kernel function $k(\cdot, \cdot)$; then write the relationship between K and $\Phi(X)$. Lastly, show that these two definitions are equivalent.

K is the $n \times n$ matrix with components $K_{ij} = k(X_i, X_j)$. Also, $K = \Phi(X)\Phi(X)^\top$.

These two characterizations are equivalent because $K_{ij} = k(X_i, X_j) = \Phi(X_i) \cdot \Phi(X_j)$ is the inner product of row i of $\Phi(X)$ and column j of $\Phi(X)^\top$, which implies that $K = \Phi(X)\Phi(X)^\top$.

(e) [2 pts] Fill in the space: the first principle component direction of the *featurized* design matrix $\Phi(X)$ is any nonzero vector $w \in \mathbb{R}^D$ that maximizes the *Rayleigh quotient*, which is .

$$\frac{w^\top \Phi(X)^\top \Phi(X) w}{w^\top w}.$$

(f) [4 pts] Show that the problem of maximizing this Rayleigh quotient is equivalent to maximizing

$$\frac{a^\top B a}{a^\top C a}$$

for some positive semidefinite matrices $B, C \in \mathbb{R}^{n \times n}$, where $a \in \mathbb{R}^n$ is a vector of dual weights. This expression is called a *generalized Rayleigh quotient*. What are the matrices B and C? For full points, express them in a form that does not require any direct computation of the feature vectors Φ, which could be extremely long.

$\frac{w^\top \Phi(X)^\top \Phi(X) w}{w^\top w} = \frac{a^\top \Phi(X)\Phi(X)^\top \Phi(X)\Phi(X)^\top a}{a^\top \Phi(X)\Phi(X)^\top a} = \frac{a^\top K^2 a}{a^\top K a}$. Hence $B = K^2$ and $C = K$.

Q6. [17 pts] Learning Mixtures of Gaussians with k-Means

Let $X_1, \ldots, X_n \in \mathbb{R}^d$ be independent, identically distributed points sampled from a mixture of two normal (Gaussian) distributions. They are drawn independently from the probability distribution function (PDF)

$$p(x) = \theta\, N_1(x) + (1-\theta)\, N_2(x), \quad \text{where } N_1(x) = \frac{1}{(\sqrt{2\pi})^d} e^{-\|x-\mu_1\|^2/2} \text{ and } N_2(x) = \frac{1}{(\sqrt{2\pi})^d} e^{-\|x-\mu_2\|^2/2}$$

are the PDFs for the isotropic multivariate normal distributions $\mathcal{N}(\mu_1, 1)$ and $\mathcal{N}(\mu_2, 1)$, respectively. The parameter $\theta \in (0, 1)$ is called the *mixture proportion*. In essence, we flip a biased coin to decide whether to draw a point from the first Gaussian (with probability θ) or the second (with probability $1 - \theta$).

Each data point is generated as follows. First draw a random Z_i, which has value 1 with probability θ, and has value 2 with probability $1 - \theta$. Then, draw $X_i \sim \mathcal{N}(\mu_{Z_i}, 1)$. Our learning algorithm gets X_i as an input, but does not know Z_i.

Our goal is to find the maximum likelihood estimates of the three unknown distribution parameters $\theta \in (0, 1)$, $\mu_1 \in \mathbb{R}^d$, and $\mu_2 \in \mathbb{R}^d$ from the sample points $X_1, \ldots, X_n$. Unlike MLE for one Gaussian, it is **not** possible to give explicit analytic formulas for these estimates. Instead, we develop a variant of k-means clustering which (often) converges to the correct maximum likelihood estimates of θ, μ_1, and μ_2. This variant doesn't assign each point entirely to one cluster; rather, each point is assigned an estimated posterior probability of coming from normal distribution 1.

(a) [4 pts] Let $\tau_i = P(Z_i = 1 | X_i)$. That is, τ_i is the posterior probability that point X_i has $Z_i = 1$. Use Bayes' Theorem to express τ_i in terms of X_i, θ, μ_1, μ_2, and the Gaussian PDFs $N_1(x)$ and $N_2(x)$. To help you with part (c), also write down a similar formula for $1 - \tau_i$, which is the posterior probability that $Z_i = 2$.

Bayes' Theorem implies that

$$\tau_i = \frac{\theta N_1(X_i)}{\theta N_1(X_i) + (1-\theta) N_2(X_i)}, \quad 1 - \tau_i = \frac{(1-\theta) N_2(X_i)}{\theta N_1(X_i) + (1-\theta) N_2(X_i)}.$$

(b) [3 pts] Write down the log-likelihood function, $\ell(\theta, \mu_1, \mu_2; X_1, \ldots, X_n) = \ln p(X_1, \ldots, X_n)$, as a summation. Note: it doesn't simplify much.

Because the samples are iid, $p(X_1, \ldots, X_n) = \prod_{i=1}^n p(X_i)$, so

$$\ell(\theta, \mu_1, \mu_2; X_1, \ldots, X_n) = \sum_{i=1}^n \ln(\theta N_1(X_i) + (1-\theta) N_2(X_i)).$$

(c) [3 pts] Express $\frac{\partial \ell}{\partial \theta}$ in terms of θ and $\tau_i, i \in \{1, \ldots, n\}$ and simplify as much as possible. There should be no normal PDFs explicitly in your solution, though the τ_i's may implicitly use them. Hint: Recall that $(\ln f(x))' = \frac{f'(x)}{f(x)}$.

$$\frac{\partial \ell}{\partial \theta} = \sum_{i=1}^n \left(\frac{\tau_i}{\theta} - \frac{1-\tau_i}{1-\theta} \right) = \frac{1}{\theta - \theta^2} \sum_{i=1}^n (\tau_i - \theta) = \frac{(\sum \tau_i) - \theta n}{\theta - \theta^2}.$$

(Any of these expressions is simple enough to receive full marks.)

(d) [4 pts] Express $\nabla_{\mu_1}\ell$ in terms of μ_1 and $\tau_i, X_i, i \in \{1,\ldots,n\}$. Do the same for $\nabla_{\mu_2}\ell$ (but in terms of μ_2 rather than μ_1). Again, there should be no normal PDFs explicitly in your solution, though the τ_i's may implicitly use them.
Hint: It will help (and get you part marks) to first write $\nabla_{\mu_1} N_1(x)$ as a function of $N_1(x)$, x, and μ_1.

$$\begin{aligned}\nabla_{\mu_1}\ell &= \sum_{i=1}^{n} \frac{\theta \nabla_{\mu_1} N_1(X_i)}{\theta N_1(X_i) + (1-\theta)N_2(X_i)} \\ &= \sum_{i=1}^{n} \frac{\theta N_1(X_i)}{\theta N_1(X_i) + (1-\theta)N_2(X_i)}(X_i - \mu_1) \\ &= \sum_{i=1}^{n} \tau_i (X_i - \mu_1).\end{aligned}$$

Similarly,

$$\nabla_{\mu_2}\ell = \sum_{i=1}^{n} (1-\tau_i)(X_i - \mu_2).$$

(e) [3 pts] We conclude: if we know μ_1, μ_2, and θ, we can compute the posteriors τ_i. On the other hand, if we know the τ_i's, we can estimate μ_1, μ_2, and θ by using the derivatives in parts (c) and (d) to find the maximum likelihood estimates. This leads to the following k-means-like algorithm.

- Initialize $\tau_1, \tau_2, \ldots, \tau_n$ to arbitrary values in the range $[0, 1]$.
- Repeat the following two steps.
  1. Update the Gaussian cluster parameters: for fixed values of $\tau_1, \tau_2, \ldots, \tau_n$, update μ_1, μ_2, and θ.
  2. Update the posterior probabilities: for fixed values of μ_1, μ_2 and θ, update $\tau_1, \tau_2, \ldots, \tau_n$.

In part (a), you wrote the update rule for step 2. Using your results from parts (c) and (d), write down the explicit update formulas for step 1.

$$\mu_1 \leftarrow \frac{\sum_{i=1}^{n} \tau_i X_i}{\sum_{i=1}^{n} \tau_i}, \quad \mu_2 \leftarrow \frac{\sum_{i=1}^{n} (1-\tau_i) X_i}{\sum_{i=1}^{n} (1-\tau_i)}, \quad \theta \leftarrow \frac{1}{n}\sum_{i=1}^{n} \tau_i.$$